

Design of Complex IIR Digital Filters using Transformed Variables

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Abstract—The design of real IIR digital filters using conformal transform variables has been extensively researched; this paper deals with the design of *complex* IIR digital filters having non-symmetric transfer functions, and directly designed using transformed variables. The transformation used is an extension of a transform used previously for continuous-time complex filters; the extension adds a bilinear function which enables the transform to be used for designing discrete-time transfer functions. Approximations having equiripple and monotonically decreasing passbands are supported. Methods for improving numerical accuracy and minimizing coefficient sensitivities are described. A high-order cascade filter example simulated using Monte Carlo analysis demonstrates almost ideal accuracy.

Index Terms—complex filters, digital filters, IIR filters, equiripple passbands, equi-ripple stopbands, cascade filters

I. INTRODUCTION

This paper presents design techniques for obtaining discrete-time IIR complex filter approximations using transformed variables.

Complex filters, that is filters that operate on complex signals having real and imaginary parts, are often used in communication systems and phased-array signal processing [1]. Complex filters can minimize image interference when frequency modulating data signals. Digital filter realizations of complex filters are often FIR; sometimes, IIR realizations are desired as they can be more efficient and frequency selective.

Real IIR discrete-time filter approximations can be designed using the bilinear transform [2]. Alternatively, real discrete-time filter-approximations can be designed using transformed variables [4][5][3][6].

Complex IIR discrete-time filters can be realized using a real low-pass prototype and frequency translating it using a complex frequency shift as is used in [7]. This paper shows how complex IIR discrete-time filter approximations can be found directly using a new conformal transform based on extending a transform used previously for continuous-time complex filters [8]. This transform translates the passband (on the $e^{j\omega}$ axis) to the complete positive imaginary axis in the transformed domain. Frequencies in the discrete-time domain between the upper passband edge and $e^{j\pi}$ are transformed to the real axis between 0 and 1; frequencies in the discrete-time domain between $e^{-j\pi}$ and the lower passband edge are transformed to the real axis between 1 and infinity. The approximation mostly occurs in the transformed domain.

Many of the routines used for designing the discrete-time complex filters are similar to those used for designing continuous-time complex filters [12] and can be obtained at [13]. Currently, the approach can be used for designing filters having elliptic equiripple passbands or monotonic passbands; approximating group delay specifications is in early stages. The stopbands need not be symmetric and can be arbitrarily specified. Transmission zeros can be moved to optimize the stopbands, and fixed transmission zeros are supported. A few examples are described.

II. A CONFORMAL TRANSFORM SUITABLE FOR DISCRETE-TIME IIR COMPLEX FILTER APPROXIMATION

Transformed variables have been used in filter approximation since the 1960's [9][10][11]. They have also been used very effectively in the design of discrete-time transfer functions for real IIR filters [3][4][5][6]. Using transformed variables was introduced for use in filter approximation for continuous-time complex filters in [8]. They were the basis of the approximation routines (for complex IIR continuous-time complex filters) in [12] which are available at [13]. To the best of the author's knowledge, this is the first time transformed variables are reported for being used in the approximation of discrete-time *complex* IIR filters.

The transform proposed in [8] is:

$$y^2 = \frac{s - j\omega_2}{s - j\omega_1}. \quad (1)$$

This transform was the basis of the procedures described in [12] for finding filter approximations for single-passband continuous-time complex filters. The procedure can be extended using a bilinear transform so it can also be used for finding approximations for *discrete-time* complex IIR filters. First, some of our terminology will be explained. In the non-transformed domain, the frequency variable is ω and we have the Laplace variable $s = j\omega$. Since we are designing discrete-time filters, we always assume the sampling time is unity, and that after the transfer functions are found, they will be frequency-scaled for applications where the sampling time is not unity; routines to do this scaling can be obtained at [13]. We also use the z -transform where $z = e^{j\omega}$. In the transformed domain, we denote the transformed variable as y , and for frequencies ω , in the non-transformed passband, we have $y = jx$ in the transformed domain. With these preliminaries, we now present the proposed conformal transform:

$$y^2 = \frac{2\frac{1-z^{-1}}{1+z^{-1}} - j\omega_2}{2\frac{1-z^{-1}}{1+z^{-1}} - j\omega_1}. \quad (2)$$

The transfer functions in the transformed domain are doubled in order and are even having only y^2 terms. Note that we have replaced s with

$$s \Rightarrow 2\frac{1-z^{-1}}{1+z^{-1}}, \quad (3)$$

where $z = e^{j\omega}$ which is a bilinear transform¹. Although we are making use of a bilinear transform in the equation for the transformed variable, the resulting filters are in general different than those obtained using classical design methods based on bilinear transforms [2][7]. Using

$$2\frac{1-z^{-1}}{1+z^{-1}} = 2\frac{e^{j\omega/2} - e^{-j\omega/2}}{e^{j\omega/2} + e^{-j\omega/2}} = 2j\tan(\omega/2) \quad (4)$$

in (2) gives

$$y^2 = \frac{j2\tan(\omega/2) - j\omega_2}{j2\tan(\omega/2) - j\omega_1} = \frac{2\tan(\omega/2) - \omega_2}{2\tan(\omega/2) - \omega_1}. \quad (5)$$

The similarity between (1) and (5) is apparent; the domain of the transformed variable y is the same. The upper passband edge, denoted wp_2 , given by

$$wp_2 = 2\arctan(\omega_2/2) \quad (6)$$

transforms to 0, and the lower passband edge, denoted wp_1 , given by

$$wp_1 = 2\arctan(\omega_1/2) \quad (7)$$

transforms to $j\infty$. The upper stopband transforms to the real axis with $e^{j\pi}$ transforming to 1, and the upper transition region next to the passband transforms to 0. The lower stopband frequency $e^{-j\pi}$ transforms to 1, and the frequency next to the passband transforms to ∞ . The right-hand side of (5) is real and negative for ω in the passband, and real and positive for ω not in the passband. The proposed $Z2Y^2$ transform is shown graphically in Fig. 1.

The transform can be extended using analytic continuation to map the unit circle and inside the unit circle, of the discrete time z -domain, to the upper-right quadrant of the y -domain; going from z to y , we have:

$$y^2 = \frac{2(z-1) - jwp_2(z+1)}{2(z-1) - jwp_1(z+1)}. \quad (8)$$

The inverse transform is:

$$z = \frac{2 + jwp_2 - y^2(2 + jwp_1)}{2 - jwp_2 - y^2(2 - jwp_1)}. \quad (9)$$

These equations can be used to transform a transfer function in z to a transfer function in y , and vice versa, by using them to transform the zeros and poles of the transfer functions. The transformed function in y will be a *polynomial* in y^2 .

Consider an example transfer function of a simple second-order discrete-time complex filter given by

¹This approach can be used in extending most continuous-time transform-based approximation algorithms for finding discrete-time approximations.

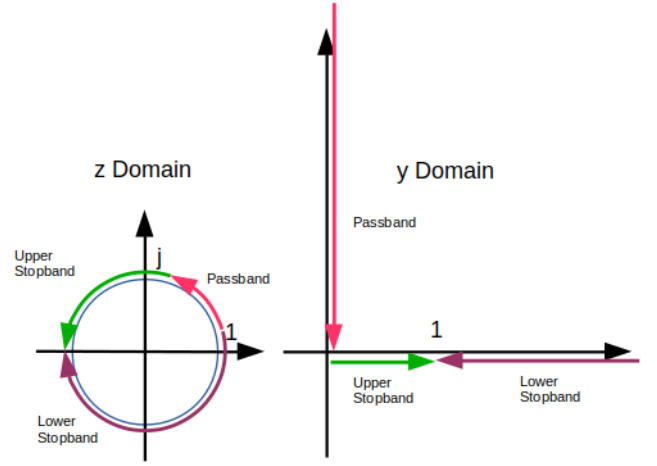


Fig. 1. Graphical representation of proposed conformal transform.

$$H_z = 0.17139 \frac{(z - (0.9847 - 0.174i))(z - (0.438 + 0.899i))}{(z - (0.8245 + 0.1981i))(z - (0.6449 + 0.5506i))}. \quad (10)$$

Its magnitude response is shown in Fig. 2.

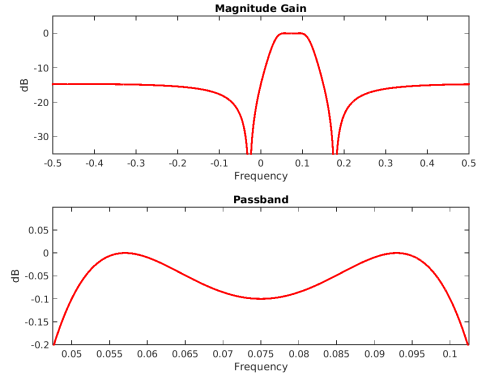


Fig. 2. The magnitude responses of the transformed passband and stopbands

Transforming the zeros and poles to the y -domain gives (ignoring the gain scaling for now):

$$H_y = \frac{(y^2 - 0.8020^2)(y^2 - 1.29495^2)}{(y^2 - (0.6249 + 0.2402j)^2)(y^2 - (1.4480 + 0.5565j)^2)}. \quad (11)$$

Note H_y is a function of y^2 and that the loss poles transform to a pair of real poles², equal in magnitude and having opposite signs. The magnitude of this transfer function has been plotted for y on the imaginary axis and y on the real axis, the results are shown in Fig. 3. The one-to-one mapping between the passbands and stopbands in the two domains is apparent. Note that 0 in the y -domain plot corresponds to the lower passband edge and (11) can be evaluated at this frequency to set the gain.

III. COMPLEX IIR DISCRETE-TIME APPROXIMATION USING $Z2Y^2$ TRANSFORM

Currently, the design routines support monotonic and elliptic passbands and arbitrary stopbands. In addition, fixed zeros can

²In classic filter design, the loss-poles are the transmission zeros, that is the zeros of H_y

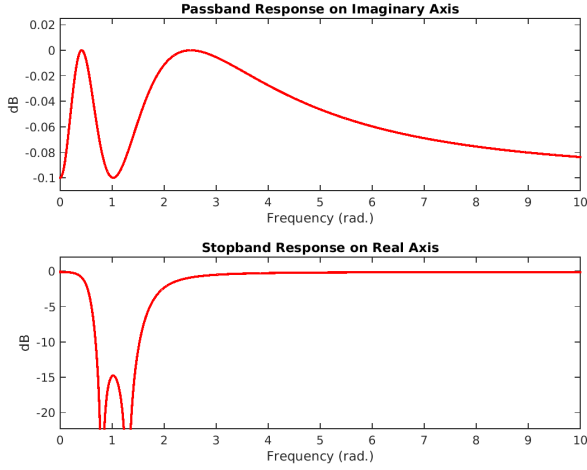


Fig. 3. The magnitude responses of the transformed passband and stopbands

be specified, perhaps at dc or half the sampling frequency, or at frequencies where unwanted images might occur. For movable loss-poles, initial guesses are given; the routines are not sensitive to these values.

The initial loss-poles guesses are transformed from the z -domain to the y -domain. The characteristic function (used in Feldtkeller's equation [11]) is then found in the y -domain. First, the loss-poles are adapted to try and equalize margins between the stopband and the stopband specifications using the modified Remez algorithm. This routine is similar (but significantly improved) to that used (and described in detail) in [12]. At each iteration, the reflection zeros are found depending on whether an elliptic or a monotonic passband is desired, as in [12]. After the characteristic function is found in the y -domain, the reflection zeros and loss-poles are transformed back to the z -domain, and then Feldtkeller's equation is used to find the desired transfer function. When solving for H (the filter transfer function) using Feldtkeller's equation for discrete-time filters, the poles inside the unit circle are chosen, as opposed to poles in the left half plane chosen when designing continuous-time filters.

It is better to solve Feldtkeller's equation in the y -domain before transforming back to the z -domain. Solving Feldtkeller's Equation in the y -domain is more complicated; this complication significantly improves numerical accuracy. Details of iteratively solving Feldtkeller's Equation in the y -domain are found in [13]. We speculate this improvement is due to: a) the passband and stopbands being expanded over the complete axes, and b) the imaginary axis for the passband being orthogonal to the real axis of the stopband.

Consider the second-order example introduced in section II. We have in z , the loss poles given by

$$P_z(z) = (z - pz_1)(z - pz_2) \quad (12)$$

which transforms to

$$P_y(y) = (y - py_1)(y + py_1)(y - py_2)(y + py_2) \quad (13)$$

where py_i is found using (2). We can now group the negative roots of P_y ³ into a new function

$$Q_y(y) = (y + py_1)(y + py_2) \quad (14)$$

and when writing Feldtkeller's expression in the y -domain, we have

$$K(y)K'(-y) = \epsilon^2 \frac{F_y(y)F_y'(-y)}{P_y(y)P_y'(-y)} = \epsilon^2 \frac{F_y(y)F_y'(-y)}{Q_y(y)Q_y'(-y)}. \quad (15)$$

We can now follow classical filter approximation procedures, for example, the procedure as described in Section 12-4 of [11] to choose $F_y(y)$. For an equiripple passband, we take

$$F_y(y) = Ev(Q_y(y)) = (Q_y(y) + Q_y'(-y))/2. \quad (16)$$

The design procedure is now:

- 1) Choose initial loss poles (in the z -domain) and transform them to the y domain
- 2) Form the characteristic function in the y -domain using (16) for elliptic filters or the approach described in [12] for monotonic filters.
- 3) Iterate on the pole positions of the movable poles in the y -domain to equalize stopband margins using the approach proposed in [12].
- 4) Repeat 2-3 until stopband margins are equalized (if due to fixed poles, the problem is over-determined, there may be one or more margins greater than the equalized margins, which is fine)
- 5) Solve for the H_y (the inverse transfer function) using Feldtkeller's equation in the y -domain.
- 6) Transform the zeros and poles of $H_y(y)H_y'(-y)$ back to the z -domain
- 7) Choose the zeros of H_z to be those inside the unit circle
- 8) Invert H_z to find the forward-gain transfer function.

The doubling in order of the Z^2Y^2 transform does not cause a degradation of numerical accuracy. The pole placement routines are based on log functions and the doubling just changes multiplication constants. When solving Feldtkeller's equation, a further $s' = s^2$ transform is used to halve the order to be the same as in the Z -domain.

IV. 13TH ORDER EXAMPLE

A 13th-order filter was designed using the procedure just outlined. Relative to a sampling frequency of 1Hz, the 0.05db passband was between 0.05Hz and 0.1Hz (an elliptic filter). The lower stopband was between -0.5Hz and 0Hz, and the upper stopband was between 0.15Hz and 0.5Hz. Two fixed poles were specified at half the sampling frequency, and a fixed pole was specified at dc. The lower stopband had six movable loss-poles, and the upper stopband had four movable loss-poles. The resulting magnitude transfer functions are shown in Fig. 4. Feldtkeller's equation was solved in the non-transformed z -domain for the transfer function shown in red; Feldtkeller's equation was solved in the transformed y -domain for the transfer function shown in blue. When solving Feldtkeller's

³We could have arbitrarily grouped the positive roots; the final results are the same.

equation (in the y -domain), Muller's algorithm was used to eliminate the need to use inaccurate polynomial addition. The latter transfer function is extremely close to ideal whereas the former is highly distorted and not usable. Although few if any applications require such large stopband attenuation (over 260dB in the lower stopband), this example shows the accuracy and robustness of the transform-based approach.

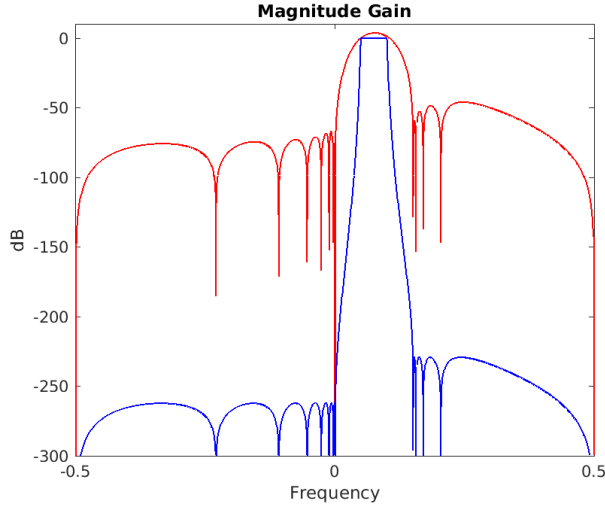


Fig. 4. High-order filters: red, traditional approach, blue, Feldtkeller's equation solved in the transformed domain using Muller algorithm.

After the transfer functions have been found, then the final complex digital IIR filter can be realized using structures such as a cascade filter; design procedures for complex cascade filters were described in [7] with Matlab code given in [13]. Based on Monte Carlo analysis, we found that *using first-order complex sections is much less sensitive to coefficient inaccuracies than using second-order sections*⁴. Two complex

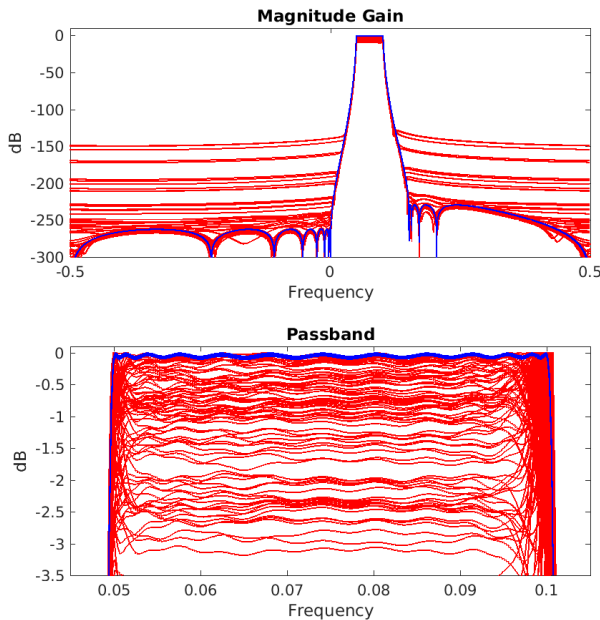


Fig. 5. FFT's of impulse responses of cascade realizations: red) using biquadratic sections and blue) using first-order sections.

⁴This is the opposite of the case for realizing *real* filters.

cascade filters were designed, and their 8192 sample impulse responses were simulated and analyzed using an FFT. Each filter was simulated 100 times with their coefficients being perturbed 0.001% rms for each simulation. The magnitude of the FFT output bins is shown in Fig. 5. The red plots are the transfer functions of the biquadratic-based filter; the blue plots are the transfer functions of the first-order-based cascade complex filters. The lower sensitivity of the first-order-based realization is significant.

The approximation routines are highly accurate for complex orders up to 17 (equivalent to a 34th-order real filter) and stopband attenuation larger than 370dB. The cascade filter simulation and analysis using an FFT of an impulse response loses accuracy for orders larger than 13.

V. CONCLUSION

The design of complex IIR digital filters based on approximations using a new conformal mapping function have been presented. An example has been described that has very large non-symmetric stopband attenuation. Guidance has been given on approaches that are significantly less sensitive to numerical ill-conditioning. The example was realized using a cascade approach and it was shown that the realization based on first-order sections was less sensitive to coefficient inaccuracies compared to that based on biquadratic sections. The routines are open source and can be downloaded from github.

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